

Probabilistic sensor fusion

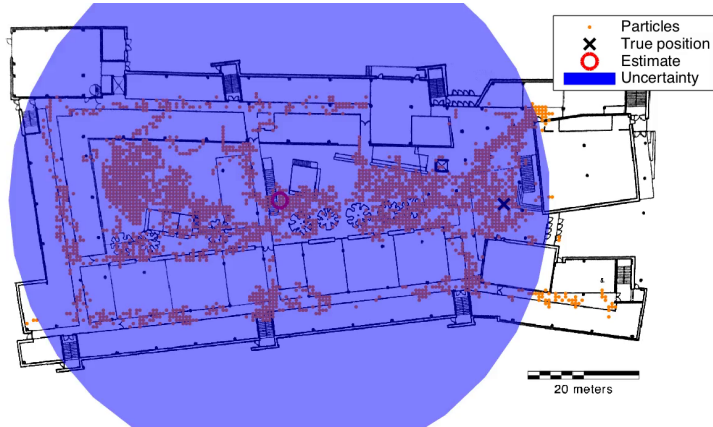
Lecture 4: Extended and unscented Kalman filters

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SC42170, EE4760

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Last lecture



Arno Solin, Simo Särkkä, Juho Kannala, and Esa Rahtu (2016). Terrain navigation in the magnetic landscape: Particle filtering for indoor positioning. Proceedings of the European Navigation Conference (ENC).

Nonlinear estimation

State space model:

$$\begin{aligned}\mathbf{x}_{k+1} &= f(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k) \\ \mathbf{y}_k &= h(\mathbf{x}_k, \mathbf{u}_k, \mathbf{v}_k)\end{aligned}$$

Goal: Estimate $\mathbf{x}_k$, $k = 1, \dots, K$ and obtain information about the uncertainty of the estimate. In other words, obtain the posterior $p(\mathbf{x}_{1:K} \mid \mathbf{y}_{1:K})$ or recursively compute $p(\mathbf{x}_k \mid \mathbf{y}_{1:k})$.

Nonlinear estimation

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Nonlinear estimation

State space model:

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k) + \mathbf{w}_k$$

$$\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$$

$$\mathbf{y}_k = h(\mathbf{x}_k) + \mathbf{v}_k$$

$$\mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$$

Goal: Estimate $\mathbf{x}_k$, $k = 1, \dots, K$ and obtain information about the uncertainty of the estimate. In other words, obtain the posterior $p(\mathbf{x}_{1:K} \mid \mathbf{y}_{1:K})$ or recursively compute $p(\mathbf{x}_k \mid \mathbf{y}_{1:k})$.

Today: Approximate posterior distribution using a Gaussian distribution.

- Recap Kalman filtering
- EKF
- UKF
- Practical aspects: Tuning and how to choose which algorithm to use

- **Recap Kalman filtering**
- EKF
- UKF
- Practical aspects: Tuning and how to choose which algorithm to use

Remember: Kalman filtering algorithm

State space model:

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{w}_k,$$

$$\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}),$$

$$\mathbf{y}_k = \mathbf{C}\mathbf{x}_k + \mathbf{v}_k,$$

$$\mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}),$$

Measurement update: With $\mathbf{K}_k = \mathbf{P}_{k|k-1}\mathbf{C}^\top (\mathbf{C}\mathbf{P}_{k|k-1}\mathbf{C}^\top + \mathbf{R})^{-1}$ being the Kalman gain,

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k (\mathbf{y}_k - \mathbf{C}\hat{\mathbf{x}}_{k|k-1})$$

$$\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1} - \mathbf{P}_{k|k-1}\mathbf{C}^\top (\mathbf{C}\mathbf{P}_{k|k-1}\mathbf{C}^\top + \mathbf{R})^{-1} \mathbf{C}\mathbf{P}_{k|k-1}$$

Time update:

$$\hat{\mathbf{x}}_{k+1|k} = \mathbf{A}\hat{\mathbf{x}}_{k|k}$$

$$\mathbf{P}_{k+1|k} = \mathbf{A}\mathbf{P}_{k|k}\mathbf{A}^\top + \mathbf{Q}$$

Additional notation

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k (\mathbf{y}_k - \mathbf{C}\hat{\mathbf{x}}_{k|k-1})$$

$$\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1} - \mathbf{P}_{k|k-1} \mathbf{C}^T \left(\mathbf{C} \mathbf{P}_{k|k-1} \mathbf{C}^T + \mathbf{R} \right)^{-1} \mathbf{C} \mathbf{P}_{k|k-1}$$

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$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k (\mathbf{y}_k - \hat{\mathbf{y}}_{k|k-1})$$

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Additional notation

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k \boldsymbol{\varepsilon}_k$$

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$$\boldsymbol{\varepsilon}_k = \mathbf{y}_k - \hat{\mathbf{y}}_{k|k-1}$$

$$\mathbb{E} [\boldsymbol{\varepsilon}_k] = \mathbb{E} [\mathbf{y}_k - \mathbf{C} \hat{\mathbf{x}}_{k|k-1}] = \mathbb{E} [\mathbf{C} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k-1}) + \mathbf{v}_k] = 0$$

$$\text{cov}(\boldsymbol{\varepsilon}_k) = \mathbb{E} [\boldsymbol{\varepsilon}_k \boldsymbol{\varepsilon}_k^T] = \mathbf{C} \mathbf{P}_{k|k-1} \mathbf{C}^T + \mathbf{R} = \mathbf{S}_k$$

- Recap Kalman filtering
- **EKF**
- UKF
- Practical aspects: Tuning and how to choose which algorithm to use

Adapting the Kalman filtering algorithm

Linear state space model:

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{w}_k,$$

$$\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}),$$

$$\mathbf{y}_k = \mathbf{C}\mathbf{x}_k + \mathbf{v}_k,$$

$$\mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}),$$

Measurement update: With $\mathbf{K}_k = \mathbf{P}_{k|k-1}\mathbf{C}^\top (\mathbf{C}\mathbf{P}_{k|k-1}\mathbf{C}^\top + \mathbf{R})^{-1}$ being the Kalman gain,

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Time update:

$$\hat{\mathbf{x}}_{k+1|k} = \mathbf{A}\hat{\mathbf{x}}_{k|k}$$

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Nonlinear state space model:

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$$\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}),$$

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Standard Kalman filter equations can't be used anymore!!

Measurement update: With $\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{C}^T (\mathbf{C} \mathbf{P}_{k|k-1} \mathbf{C}^T + \mathbf{R})^{-1}$ being the Kalman gain,

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Adapting the Kalman filtering algorithm

Approximate state space model as linear? (bad model!!):

$$\mathbf{x}_{k+1} \approx \mathbf{F}\mathbf{x}_k + \mathbf{w}_k,$$

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Measurement update: With $\mathbf{K}_k = \mathbf{P}_{k|k-1}\mathbf{C}^\top (\mathbf{C}\mathbf{P}_{k|k-1}\mathbf{C}^\top + \mathbf{R})^{-1}$ being the Kalman gain,

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Measurement update: With $\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}^\top (\mathbf{H} \mathbf{P}_{k|k-1} \mathbf{H}^\top + \mathbf{R})^{-1}$ being the Kalman gain,

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$$\hat{\mathbf{x}}_{k+1|k} = \mathbf{F}\hat{\mathbf{x}}_{k|k}$$

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Adapting the Kalman filtering algorithm

Approximate state space model as time-varying linear:

$$\mathbf{x}_{k+1} \approx \mathbf{F}_k \mathbf{x}_k + \mathbf{w}_k,$$

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Adapting the Kalman filtering algorithm

Approximate state space model as time-varying linear:

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Measurement update: With $\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^T (\mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^T + \mathbf{R})^{-1}$ being the Kalman gain,

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Approximate state space model as time-varying linear:

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Measurement update: With $\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^T (\mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^T + \mathbf{R})^{-1}$ being the Kalman gain,

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Adapting the Kalman filtering algorithm

Approximate state space model as time-varying linear:

$$\mathbf{x}_{k+1} \approx \mathbf{F}_k \mathbf{x}_k + \mathbf{w}_k,$$

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$$\mathbf{y}_k \approx \mathbf{H}_k \mathbf{x}_k + \mathbf{v}_k,$$

$$\mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}),$$

But we can do even better!

Measurement update: With $\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^T (\mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^T + \mathbf{R})^{-1}$ being the Kalman gain,

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Approximate state space model as time-varying linear:

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Time update:

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Adapting the Kalman filtering algorithm

Approximate state space model as time-varying linear:

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Adapting the Kalman filtering algorithm

Approximate state space model as time-varying linear:

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$$\mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}),$$

This is the extended Kalman filter!

Measurement update: With $\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^T (\mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^T + \mathbf{R})^{-1}$ being the Kalman gain,

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Computing the matrix $\mathbf{F}_k$

Linearize the dynamic model using a first order Taylor expansion around some $\bar{\mathbf{x}}$

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k) + \mathbf{w}_k \approx f(\bar{\mathbf{x}}) + \left. \frac{df(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k=\bar{\mathbf{x}}} (\mathbf{x}_k - \bar{\mathbf{x}}) + \text{h.o.t.} + \mathbf{w}_k$$

Computing the matrix $\mathbf{F}_k$

Linearize the dynamic model using a first order Taylor expansion around $\hat{\mathbf{x}}_{k|k}$

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k) + \mathbf{w}_k \approx f(\hat{\mathbf{x}}_{k|k}) + \left. \frac{df(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k = \hat{\mathbf{x}}_{k|k}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k}) + \text{h.o.t.} + \mathbf{w}_k$$

Computing the matrix $\mathbf{F}_k$

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$$\hat{\mathbf{x}}_{k+1|k} \approx \mathbb{E}[f(\hat{\mathbf{x}}_{k|k})] + \mathbb{E} \left[\left. \frac{df(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k = \hat{\mathbf{x}}_{k|k}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k}) \right] + \mathbb{E}[\mathbf{w}_k]$$

Computing the matrix $\mathbf{F}_k$

Linearize the dynamic model using a first order Taylor expansion around some $\bar{\mathbf{x}}$

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k) + \mathbf{w}_k \approx f(\hat{\mathbf{x}}_{k|k}) + \left. \frac{df(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k = \hat{\mathbf{x}}_{k|k}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k}) + \text{h.o.t.} + \mathbf{w}_k$$

$$\hat{\mathbf{x}}_{k+1|k} \approx \mathbb{E}[f(\hat{\mathbf{x}}_{k|k})] + \mathbb{E} \left[\left. \frac{df(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k = \hat{\mathbf{x}}_{k|k}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k}) \right] + \mathbb{E}[\mathbf{w}_k]$$

Computing the matrix $\mathbf{F}_k$

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$$\hat{\mathbf{x}}_{k+1|k} \approx f(\hat{\mathbf{x}}_{k|k}) + \mathbb{E} \left[\left. \frac{df(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k = \hat{\mathbf{x}}_{k|k}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k}) \right]$$

Computing the matrix $\mathbf{F}_k$

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$$\hat{\mathbf{x}}_{k+1|k} \approx f(\hat{\mathbf{x}}_{k|k}) + \mathbb{E} \left[\left. \frac{df(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k = \hat{\mathbf{x}}_{k|k}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k}) \right]$$

Computing the matrix $\mathbf{F}_k$

Linearize the dynamic model using a first order Taylor expansion around some $\bar{\mathbf{x}}$

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k) + \mathbf{w}_k \approx f(\hat{\mathbf{x}}_{k|k}) + \left. \frac{df(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k = \hat{\mathbf{x}}_{k|k}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k}) + \text{h.o.t.} + \mathbf{w}_k$$

$$\hat{\mathbf{x}}_{k+1|k} \approx f(\hat{\mathbf{x}}_{k|k})$$

Computing the matrix $\mathbf{F}_k$

Linearize the dynamic model using a first order Taylor expansion around some $\bar{\mathbf{x}}$

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k) + \mathbf{w}_k \approx f(\hat{\mathbf{x}}_{k|k}) + \left. \frac{df(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k = \hat{\mathbf{x}}_{k|k}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k}) + \text{h.o.t.} + \mathbf{w}_k$$

$$\hat{\mathbf{x}}_{k+1|k} \approx f(\hat{\mathbf{x}}_{k|k})$$

$$\mathbb{E}[(\mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1|k})(\mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1|k})^T]$$

Computing the matrix $\mathbf{F}_k$

Linearize the dynamic model using a first order Taylor expansion around some $\bar{\mathbf{x}}$

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k) + \mathbf{w}_k \approx f(\hat{\mathbf{x}}_{k|k}) + \left. \frac{df(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k = \hat{\mathbf{x}}_{k|k}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k}) + \text{h.o.t.} + \mathbf{w}_k$$

$$\hat{\mathbf{x}}_{k+1|k} \approx f(\hat{\mathbf{x}}_{k|k})$$

$$\begin{aligned} & \mathbb{E} [(\mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1|k})(\mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1|k})^T] \\ &= \mathbb{E} \left[\left(f(\hat{\mathbf{x}}_{k|k}) + \left. \frac{df(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k = \hat{\mathbf{x}}_{k|k}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k}) + \mathbf{w}_k - f(\hat{\mathbf{x}}_{k|k}) \right) (\cdot)^T \right] \end{aligned}$$

Computing the matrix $\mathbf{F}_k$

Linearize the dynamic model using a first order Taylor expansion around some $\bar{\mathbf{x}}$

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k) + \mathbf{w}_k \approx f(\hat{\mathbf{x}}_{k|k}) + \left. \frac{df(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k = \hat{\mathbf{x}}_{k|k}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k}) + \text{h.o.t.} + \mathbf{w}_k$$

$$\hat{\mathbf{x}}_{k+1|k} \approx f(\hat{\mathbf{x}}_{k|k})$$

$$\begin{aligned} & \mathbb{E} [(\mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1|k})(\mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1|k})^T] \\ &= \mathbb{E} \left[\left(f(\hat{\mathbf{x}}_{k|k}) + \left. \frac{df(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k = \hat{\mathbf{x}}_{k|k}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k}) + \mathbf{w}_k - f(\hat{\mathbf{x}}_{k|k}) \right) (\cdot)^T \right] \\ &= \mathbb{E} \left[\left(\left. \frac{df(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k = \hat{\mathbf{x}}_{k|k}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k}) + \mathbf{w}_k \right) (\cdot)^T \right] \end{aligned}$$

Computing the matrix $\mathbf{F}_k$

Linearize the dynamic model using a first order Taylor expansion around some $\bar{\mathbf{x}}$

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k) + \mathbf{w}_k \approx f(\hat{\mathbf{x}}_{k|k}) + \left. \frac{df(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k = \hat{\mathbf{x}}_{k|k}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k}) + \text{h.o.t.} + \mathbf{w}_k$$

$$\hat{\mathbf{x}}_{k+1|k} \approx f(\hat{\mathbf{x}}_{k|k})$$

$$\begin{aligned} & \mathbb{E}[(\mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1|k})(\mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1|k})^T] \\ &= \mathbb{E} \left[\left(f(\hat{\mathbf{x}}_{k|k}) + \left. \frac{df(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k = \hat{\mathbf{x}}_{k|k}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k}) + \mathbf{w}_k - f(\hat{\mathbf{x}}_{k|k}) \right) (\cdot)^T \right] \\ &= \mathbb{E} \left[\left(\left. \frac{df(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k = \hat{\mathbf{x}}_{k|k}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k}) + \mathbf{w}_k \right) (\cdot)^T \right] \\ &= \mathbf{F}_k \mathbf{P}_{k|k} \mathbf{F}_k^T + \mathbf{Q} \end{aligned}$$

Extended Kalman filtering

State space model:

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k) + \mathbf{w}_k,$$

$$\mathbf{y}_k = h(\mathbf{x}_k) + \mathbf{v}_k,$$

$$\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}),$$

$$\mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}),$$

Time update: With $\mathbf{F}_k = \left. \frac{df(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k = \hat{\mathbf{x}}_{k|k}}$,

$$\hat{\mathbf{x}}_{k+1|k} = f(\hat{\mathbf{x}}_{k|k})$$

$$\mathbf{P}_{k+1|k} = \mathbf{F}_k \mathbf{P}_{k|k} \mathbf{F}_k^T + \mathbf{Q}$$

Computing the matrix $\mathbf{H}_k$

Linearize the measurement model using a first order Taylor expansion around some $\bar{\mathbf{x}}$

$$\mathbf{y}_k = h(\mathbf{x}_k) + \mathbf{v}_k \approx h(\bar{\mathbf{x}}) + \left. \frac{dh(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k=\bar{\mathbf{x}}} (\mathbf{x}_k - \bar{\mathbf{x}}) + \text{h.o.t.} + \mathbf{v}_k$$

Computing the matrix $\mathbf{H}_k$

Linearize the measurement model using a first order Taylor expansion around $\hat{\mathbf{x}}_{k|k-1}$

$$\mathbf{y}_k = h(\mathbf{x}_k) + \mathbf{v}_k \approx h(\hat{\mathbf{x}}_{k|k-1}) + \left. \frac{dh(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k=\hat{\mathbf{x}}_{k|k-1}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k-1}) + \text{h.o.t.} + \mathbf{v}_k$$

Computing the matrix $\mathbf{H}_k$

Linearize the measurement model using a first order Taylor expansion around some $\bar{\mathbf{x}}$

$$\mathbf{y}_k = h(\mathbf{x}_k) + \mathbf{v}_k \approx h(\hat{\mathbf{x}}_{k|k-1}) + \left. \frac{dh(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k=\hat{\mathbf{x}}_{k|k-1}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k-1}) + \text{h.o.t.} + \mathbf{v}_k$$

$$\hat{\mathbf{y}}_{k|k-1} \approx \mathbb{E} [h(\hat{\mathbf{x}}_{k|k-1})] + \mathbb{E} \left[\left. \frac{dh(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k=\hat{\mathbf{x}}_{k|k-1}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k-1}) \right] + \mathbb{E} [\mathbf{v}_k]$$

Computing the matrix $\mathbf{H}_k$

Linearize the measurement model using a first order Taylor expansion around some $\bar{\mathbf{x}}$

$$\mathbf{y}_k = h(\mathbf{x}_k) + \mathbf{v}_k \approx h(\hat{\mathbf{x}}_{k|k-1}) + \left. \frac{dh(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k=\hat{\mathbf{x}}_{k|k-1}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k-1}) + \text{h.o.t.} + \mathbf{v}_k$$

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Computing the matrix $\mathbf{H}_k$

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Computing the matrix $\mathbf{H}_k$

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Computing the matrix $\mathbf{H}_k$

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Computing the matrix $\mathbf{H}_k$

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$$\hat{\mathbf{y}}_{k|k-1} \approx h(\hat{\mathbf{x}}_{k|k-1})$$

Computing the matrix $\mathbf{H}_k$

Linearize the measurement model using a first order Taylor expansion around some $\bar{\mathbf{x}}$

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$$\hat{\mathbf{y}}_{k|k-1} \approx h(\hat{\mathbf{x}}_{k|k-1})$$

$$\mathbb{E}[(\mathbf{y}_k - \hat{\mathbf{y}}_{k|k-1})(\mathbf{y}_k - \hat{\mathbf{y}}_{k|k-1})^T]$$

Computing the matrix $\mathbf{H}_k$

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$$\hat{\mathbf{y}}_{k|k-1} \approx h(\hat{\mathbf{x}}_{k|k-1})$$

$$\begin{aligned} & \mathbb{E}[(\mathbf{y}_k - \hat{\mathbf{y}}_{k|k-1})(\mathbf{y}_k - \hat{\mathbf{y}}_{k|k-1})^T] \\ &= \mathbb{E} \left[\left(h(\hat{\mathbf{x}}_{k|k-1}) + \left. \frac{dh(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k=\hat{\mathbf{x}}_{k|k-1}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k-1}) + \mathbf{v}_k - h(\hat{\mathbf{x}}_{k|k-1}) \right) (\cdot)^T \right] \end{aligned}$$

Computing the matrix $\mathbf{H}_k$

Linearize the measurement model using a first order Taylor expansion around some $\bar{\mathbf{x}}$

$$\mathbf{y}_k = h(\mathbf{x}_k) + \mathbf{v}_k \approx h(\hat{\mathbf{x}}_{k|k-1}) + \left. \frac{dh(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k=\hat{\mathbf{x}}_{k|k-1}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k-1}) + \text{h.o.t.} + \mathbf{v}_k$$

$$\hat{\mathbf{y}}_{k|k-1} \approx h(\hat{\mathbf{x}}_{k|k-1})$$

$$\begin{aligned} & \mathbb{E}[(\mathbf{y}_k - \hat{\mathbf{y}}_{k|k-1})(\mathbf{y}_k - \hat{\mathbf{y}}_{k|k-1})^T] \\ &= \mathbb{E} \left[\left(h(\hat{\mathbf{x}}_{k|k-1}) + \left. \frac{dh(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k=\hat{\mathbf{x}}_{k|k-1}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k-1}) + \mathbf{v}_k - h(\hat{\mathbf{x}}_{k|k-1}) \right) (\cdot)^T \right] \\ &= \mathbb{E} \left[\left(\left. \frac{dh(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k=\hat{\mathbf{x}}_{k|k-1}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k-1}) + \mathbf{v}_k \right) (\cdot)^T \right] \end{aligned}$$

Computing the matrix $\mathbf{H}_k$

Linearize the measurement model using a first order Taylor expansion around some $\bar{\mathbf{x}}$

$$\mathbf{y}_k = h(\mathbf{x}_k) + \mathbf{v}_k \approx h(\hat{\mathbf{x}}_{k|k-1}) + \left. \frac{dh(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k=\hat{\mathbf{x}}_{k|k-1}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k-1}) + \text{h.o.t.} + \mathbf{v}_k$$

$$\hat{\mathbf{y}}_{k|k-1} \approx h(\hat{\mathbf{x}}_{k|k-1})$$

$$\begin{aligned} & \mathbb{E}[(\mathbf{y}_k - \hat{\mathbf{y}}_{k|k-1})(\mathbf{y}_k - \hat{\mathbf{y}}_{k|k-1})^T] \\ &= \mathbb{E} \left[\left(h(\hat{\mathbf{x}}_{k|k-1}) + \left. \frac{dh(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k=\hat{\mathbf{x}}_{k|k-1}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k-1}) + \mathbf{v}_k - h(\hat{\mathbf{x}}_{k|k-1}) \right) (\cdot)^T \right] \\ &= \mathbb{E} \left[\left(\left. \frac{dh(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k=\hat{\mathbf{x}}_{k|k-1}} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k-1}) + \mathbf{v}_k \right) (\cdot)^T \right] \\ &= \mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^T + \mathbf{R} \end{aligned}$$

Extended Kalman filtering

State space model:

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k) + \mathbf{w}_k,$$

$$\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}),$$

$$\mathbf{y}_k = h(\mathbf{x}_k) + \mathbf{v}_k,$$

$$\mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}),$$

Measurement update: With $\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^T (\mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^T + \mathbf{R})^{-1}$ being the Kalman gain and $\mathbf{H}_k = \left. \frac{dh(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k = \hat{\mathbf{x}}_{k|k-1}}$,

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k (\mathbf{y}_k - h(\hat{\mathbf{x}}_{k|k-1}))$$

$$\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1} - \mathbf{P}_{k|k-1} \mathbf{H}_k^T (\mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^T + \mathbf{R})^{-1} \mathbf{H}_k \mathbf{P}_{k|k-1}$$

Time update: With $\mathbf{F}_k = \left. \frac{df(\mathbf{x}_k)}{d\mathbf{x}_k} \right|_{\mathbf{x}_k = \hat{\mathbf{x}}_{k|k}}$,

$$\hat{\mathbf{x}}_{k+1|k} = f(\hat{\mathbf{x}}_{k|k})$$

$$\mathbf{P}_{k+1|k} = \mathbf{F}_k \mathbf{P}_{k|k} \mathbf{F}_k^T + \mathbf{Q}$$

Variations of the EKF

- Iterated EKF
- Second order EKF
- Smoothing

For more information see e.g.

- Simo Särkkä, Bayesian Filtering and Smoothing, Cambridge University Press, 2013.
- Fredrik Gustafsson, Statistical Sensor Fusion, Studentlitteratur, 2012.

- Recap Kalman filtering
- EKF
- **UKF**
- Practical aspects: Tuning and how to choose which algorithm to use

Unscented Kalman filter

The UKF

- Recursively approximates the mean and covariance of the predictive and filtering pdfs.
- Uses “samples” like a particle filter but choose them deterministically! These samples are called sigma points.
- Uses Gaussian approximations like a Kalman filter but computes this by directly approximating the mean and covariance rather than approximating the nonlinear functions in the state space model.
- Does not require computation of Jacobians.
- Is able to capture higher order moments caused by the nonlinear functions better than the EKF.

Unscented Transform (UT)

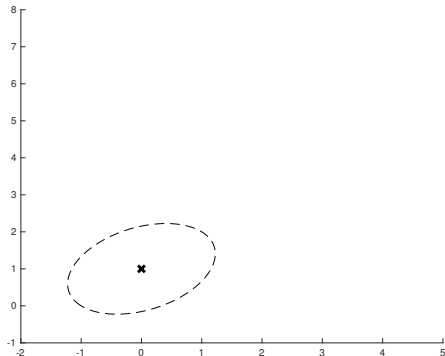
Assume: $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{P})$ and apply a nonlinear function $\mathbf{y} = f(\mathbf{x})$.

Goal: Approximate $\mathbf{y}$ using a Gaussian.

Unscented Transform (UT)

Assume: $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{P})$ and apply a nonlinear function $\mathbf{y} = f(\mathbf{x})$.

Goal: Approximate $\mathbf{y}$ using a Gaussian.

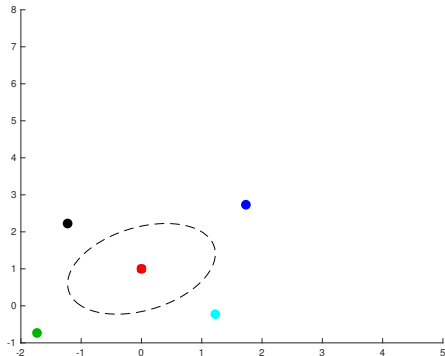


$$\mathbf{x} \in \mathbb{R}^2, \quad f(\mathbf{x}) = (x_1^2 \quad x_2^2)^T$$

Unscented Transform (UT)

Assume: $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{P})$ and apply a nonlinear function $\mathbf{y} = f(\mathbf{x})$.

Goal: Approximate $\mathbf{y}$ using a Gaussian.

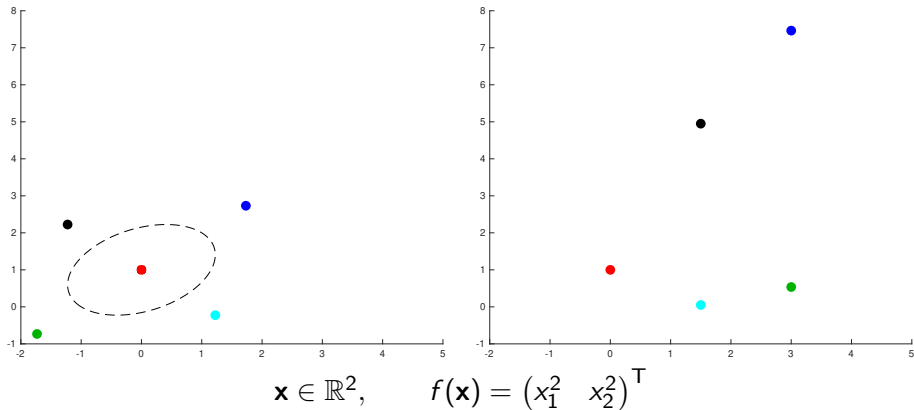


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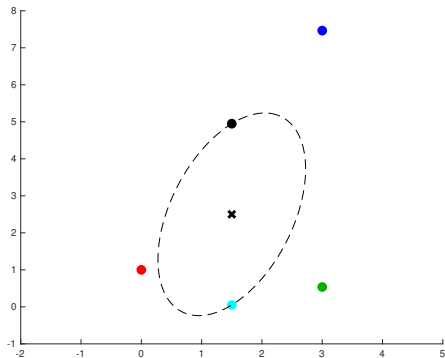
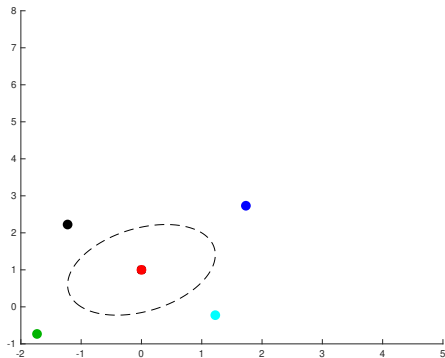
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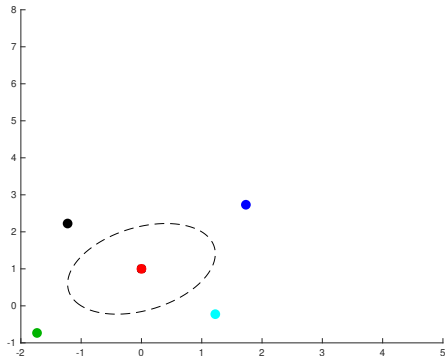
$$\boldsymbol{\chi}^{(0)} = \boldsymbol{\mu}_x, \quad \boldsymbol{\chi}^{(\pm i)} = \boldsymbol{\mu}_x \pm \sqrt{n + \lambda} [\sqrt{\mathbf{P}_x}]_i, \quad i = 1, \dots, n,$$

where $\mathbf{P} = \sqrt{\mathbf{P}} \sqrt{\mathbf{P}}^T$ and $[\sqrt{\mathbf{P}}]_i$ denotes the i 'th column of the matrix $\sqrt{\mathbf{P}}$. Also choose two sets of corresponding weights w_m, w_c .

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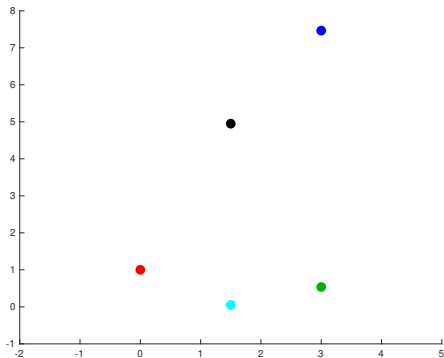
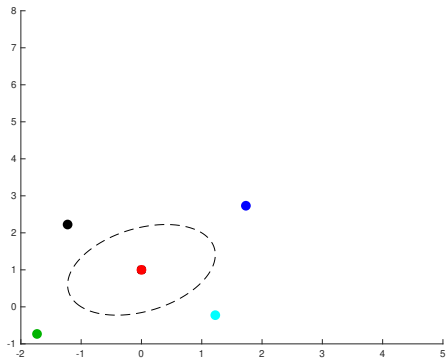
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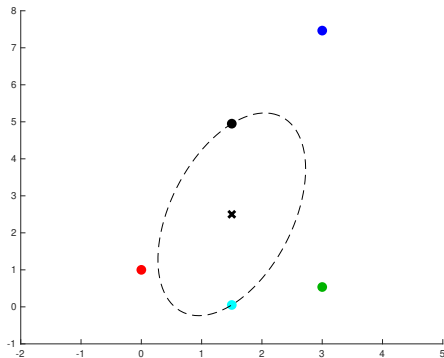
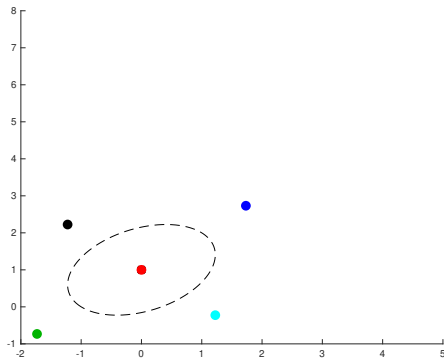
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Note also that $\mathbf{P}_{xy} = \sum_{i=-n}^n w_c^{(i)} (\boldsymbol{\chi}^{(i)} - \boldsymbol{\mu}_x) (\boldsymbol{\mathcal{Y}}^{(i)} - \boldsymbol{\mu}_y)^T$.

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Theorems 1 / 3: Joint and conditional distribution of Gaussian variables

If random variables $\mathbf{x}$ and $\mathbf{y}$ have the Gaussian probability distributions

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and the conditional distribution of x given y is given by

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and the conditional distribution of $\mathbf{x}$ given $\mathbf{y}$ is given by

$$\begin{aligned}p(\mathbf{x} | \mathbf{y}) &= \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_x + \mathbf{K}(\mathbf{y} - \mathbf{H}\boldsymbol{\mu}_x), \\ &\quad \mathbf{P}_x - \mathbf{K}\mathbf{S}^{-1}\mathbf{K}^T)\end{aligned}$$

with $\mathbf{K} = \mathbf{P}_x\mathbf{H}^T\mathbf{S}^{-1}$.

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$$\hat{\mathbf{y}}_{k|k-1} = \sum_{i=-n}^n w_m^{(i)} \mathbf{y}^{(i)}, \quad \bar{\mathbf{C}}_k = \sum_{i=-n}^n w_c^{(i)} \left(\chi^{(i)} - \hat{\mathbf{x}}_{k|k-1} \right) \left(\mathbf{y}^{(i)} - \hat{\mathbf{y}}_{k|k-1} \right)^T$$

$$\mathbf{S}_k = \sum_{i=-n}^n w_c^{(i)} \left(\mathbf{y}^{(i)} - \hat{\mathbf{y}}_{k|k-1} \right) \left(\mathbf{y}^{(i)} - \hat{\mathbf{y}}_{k|k-1} \right)^T + \mathbf{R},$$

- 4 Using $\mathbf{K}_k = \bar{\mathbf{C}}_k \mathbf{S}_k^{-1}$, update the state and covariance as

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k \left(\mathbf{y}_k - \hat{\mathbf{y}}_{k|k-1} \right), \quad \mathbf{P}_{k|k} = \mathbf{P}_{k|k-1} - \mathbf{K}_k \mathbf{S}_k \mathbf{K}_k^T.$$

Parameters of UT

- Distance between sigma points?
- Weights?

Unscented Transform (UT)

Assume: $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}_x, \mathbf{P}_x) \in \mathbb{R}^n$ and apply a nonlinear function $\mathbf{y} = f(\mathbf{x})$.

Goal: Approximate $\mathbf{y}$ using a Gaussian.

Approach:

- 1 Choose $2n + 1$ sigma points $\boldsymbol{\chi}$ as

$$\boldsymbol{\chi}^{(0)} = \boldsymbol{\mu}_x, \quad \boldsymbol{\chi}^{(\pm i)} = \boldsymbol{\mu}_x \pm \sqrt{n + \lambda} [\sqrt{\mathbf{P}_x}]_i, \quad i = 1, \dots, n,$$

where $\mathbf{P} = \sqrt{\mathbf{P}} \sqrt{\mathbf{P}}^T$ and $[\sqrt{\mathbf{P}}]_i$ denotes the i 'th column of the matrix $\sqrt{\mathbf{P}}$. Also choose two sets of corresponding weights w_m, w_c .

- 2 Propagate the sigma points through $f(\cdot)$ to obtain $\boldsymbol{\mathcal{Y}}^{(i)} = f(\boldsymbol{\chi}^{(i)})$.
- 3 Compute the mean and covariance of the transformed variable as

$$\mathbf{y} \sim \mathcal{N}(\boldsymbol{\mu}_y, \mathbf{P}_y) \quad \text{with}$$

$$\boldsymbol{\mu}_y = \sum_{i=-n}^n w_m^{(i)} \boldsymbol{\mathcal{Y}}^{(i)}, \quad \mathbf{P}_y = \sum_{i=-n}^n w_c^{(i)} (\boldsymbol{\mathcal{Y}}^{(i)} - \boldsymbol{\mu}_y) (\boldsymbol{\mathcal{Y}}^{(i)} - \boldsymbol{\mu}_y)^T$$

Note also that $\mathbf{P}_{xy} = \sum_{i=-n}^n w_c^{(i)} (\boldsymbol{\chi}^{(i)} - \boldsymbol{\mu}_x) (\boldsymbol{\mathcal{Y}}^{(i)} - \boldsymbol{\mu}_y)^T$.

Parameters of UT

- Distance between sigma points?
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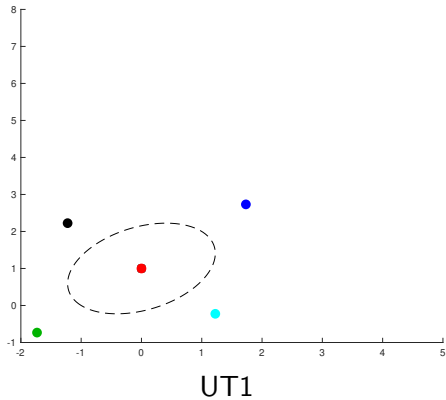
Parameter	UT1	UT2
α	$\sqrt{3/n}$	10^{-3}
κ	0	0

Parameters of UT

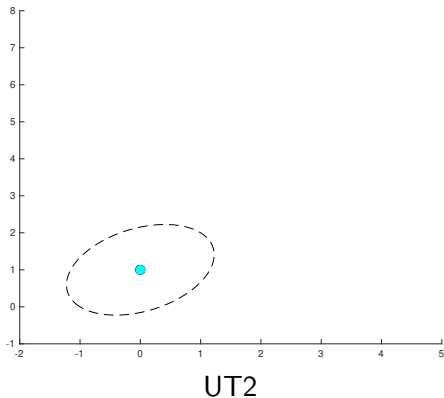
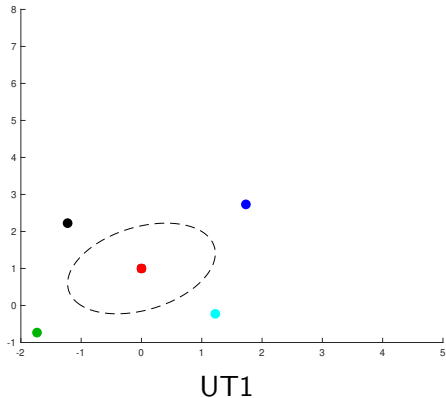
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Parameter	UT1	UT2
α	$\sqrt{3/n}$	10^{-3}
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λ	$3 - n$	$10^{-6}n - n$
$\sqrt{n + \lambda}$	$\sqrt{3}$	$10^{-3}\sqrt{n}$

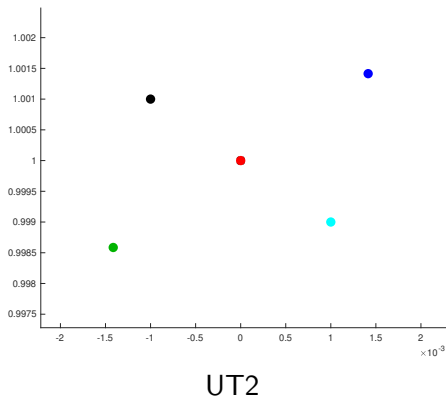
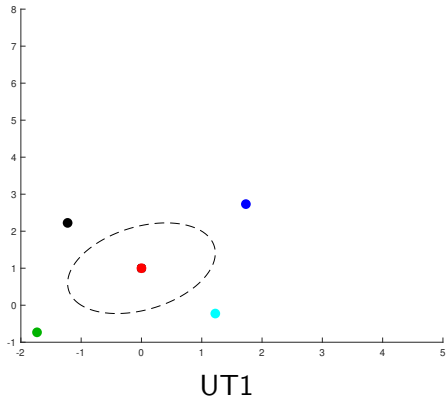
Location sigma points UT1 vs UT2



Location sigma points UT1 vs UT2



Location sigma points UT1 vs UT2



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$$w_m^{(0)} = \frac{\lambda}{n + \lambda},$$

$$w_c^{(0)} = \frac{\lambda}{n + \lambda} + (1 - \alpha^2 + \beta),$$

$$w_m^{(\pm i)} = \frac{1}{2(n + \lambda)}$$

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α	$\sqrt{3/n}$	10^{-3}
κ	0	0
β	$3/n - 1$	2
λ	$3 - n$	$10^{-6}n - n$
$\sqrt{n + \lambda}$	$\sqrt{3}$	$10^{-3}\sqrt{n}$
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Note that $\sum_{i=-n}^n w_m^{(i)} = 1$ but $\sum_{i=-n}^n w_c^{(i)} \geq 1$.

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Parameters typically used as standard but their choice can matter a lot!

ut.m

Variations of the UKF

- Different number / placement of sigma points.
- Smoothing.

For more information see e.g.

- S. J. Julier, J. K. Uhlmann, and H. F. Durrant-Whyte, A new approach for filtering nonlinear systems, Proceedings of American Control Conference, 1995.
- E. A. Wan, and R. Van Der Merwe, The unscented Kalman filter for nonlinear estimation. Proceedings of the Adaptive Systems for Signal Processing, Communications, and Control Symposium, 2000.
- Simo Särkkä, Bayesian Filtering and Smoothing, Cambridge University Press, 2013.

- Recap Kalman filtering
- EKF
- UKF
- **Practical aspects: Tuning and how to choose which algorithm to use**

Choosing between the different filters

In principle PF best captures nonlinearity, then UKF, then EKF. But:

- An EKF / UKF can work better than a PF in the case of finite number of particles and moderate nonlinearity (and they will also be faster).
- UKF typically performs better than or equal to an EKF.
- The performance of a UKF depends on the placement of the sigma points.

Unscented Kalman Filters (1)

“Any set of sigma points that encodes the mean and covariance correctly calculates the projected mean and covariance correctly to the second order.” [1]

[1] S. J. Julier and J. K. Uhlmann, Unscented filtering and nonlinear estimation, Proceedings of the IEEE, 2004.

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A simple counter-example to this statement is, however, [2], the transformation $g(\mathbf{x}) = \mathbf{x}^T \mathbf{x}$ of a variable $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathcal{I}_n)$. The posterior is known to be $\chi^2(n)$ with mean n and variance $2n$.

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n	1	2	3	4	5
“EKF1”	$\mathcal{N}(0, 0)$	$\mathcal{N}(0, 0)$	$\mathcal{N}(0, 0)$	$\mathcal{N}(0, 0)$	$\mathcal{N}(0, 0)$
“EKF2”	$\mathcal{N}(1, 2)$	$\mathcal{N}(2, 4)$	$\mathcal{N}(3, 6)$	$\mathcal{N}(4, 8)$	$\mathcal{N}(5, 10)$
UT1	$\mathcal{N}(1, 2)$	$\mathcal{N}(2, 2)$	$\mathcal{N}(3, 0)$	$\mathcal{N}(4, -4)$	$\mathcal{N}(5, -10)$
UT2	$\mathcal{N}(1, 2)$	$\mathcal{N}(2, 8)$	$\mathcal{N}(3, 18)$	$\mathcal{N}(4, 32)$	$\mathcal{N}(5, 50)$

[1] S. J. Julier and J. K. Uhlmann, Unscented filtering and nonlinear estimation, Proceedings of the IEEE, 2004.

[2] F. Gustafsson and G. Hendeby, Some relations between extended and unscented Kalman filters. IEEE Transactions on Signal Processing, 2011.

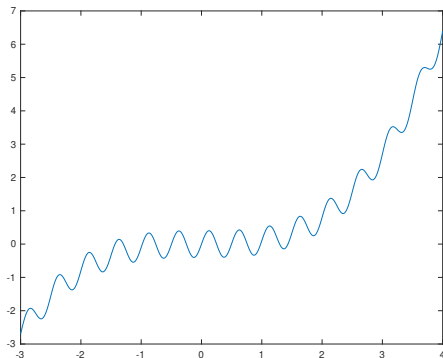
Unscented Kalman Filters (2)

Imagine localization using a Lidar sensor, which measures the distance to a non-flat wall. For localization you are interested in the direction of the wall, but not in the local variations.

- [1] K. Nielsen, C. Svahn, H. Rodriguez-Deniz and G. Hendeby, UKF parameter tuning for local variation smoothing, Proceedings of the International Conference on Multisensor Fusion and Integration for Intelligent Systems, 2021.

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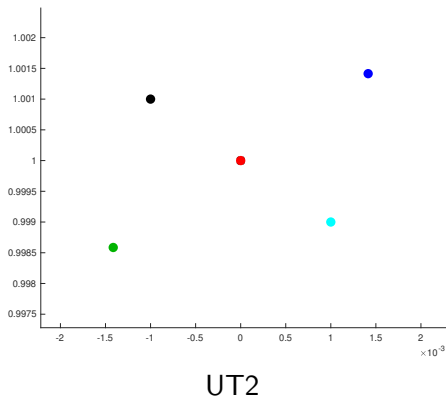
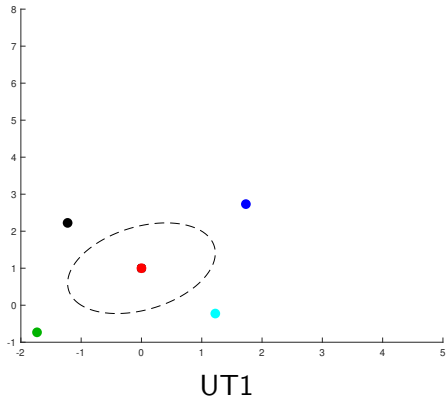
Imagine localization using a Lidar sensor, which measures the distance to a non-flat wall. For localization you are interested in the direction of the wall, but not in the local variations.

$x \sim$	$\mathcal{N}(0, 0.25^2)$		$\mathcal{N}(2.2, 0.25^2)$	
Method	μ_y	P_y	μ_y	P_y
UT1	0	0.028	1.32	0.26
UT2	0	1.58	0.18	2.93
MCT	$5 \cdot 10^{-4}$	0.08	1.11	0.21

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Location sigma points UT1 vs UT2



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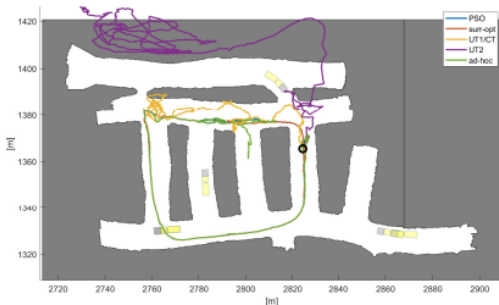
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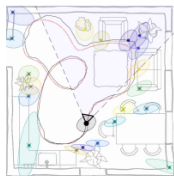


One can even optimize these values (but this is almost never done).

[1] K. Nielsen, C. Svahn, H. Rodriguez-Deniz and G. Hendeby, UKF parameter tuning for local variation smoothing,

Proceedings of the International Conference on Multisensor Fusion and Integration for Intelligent Systems, 2021.

Unscented Kalman Filters (3)

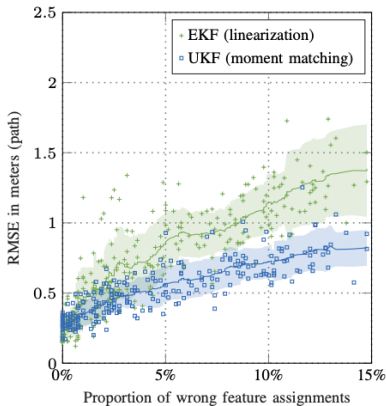


EKF step 130

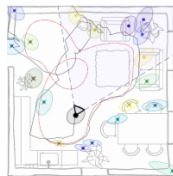


UKF step 130

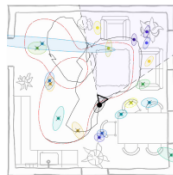
(a) No wrong assignments



(b) Normalized RMSE in path



EKF step 130



UKF step 130

(d) 15% wrong assignments

[1] A. Solin, R. Li and A. Pilzer, A look at improving robustness in visual-inertial SLAM by moment matching, Proceedings of the International Conference on Information Fusion, 2022.

Modeling and tuning

Modeling

- Performance of the algorithms highly depends on the model assumptions that are used.
- Often filters can work a bit better if the covariances are chosen a bit larger than you'd expect based on measurement noise. However, for the UKF it's better to not choose the initial covariance too large [1].

Tuning

- Use physics as much as possible (determine measurement noise covariances!). Tune typically in orders of 10.

[1] F. Gustafsson and G. Hendeby, Some relations between extended and unscented Kalman filters. IEEE Transactions on Signal Processing, 2011.

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